CANDIDATE'S DECLARATION

I hereby certify that the work which is being presented in the thesis, entitled "**A FRAMEWORK FOR DESIGNING AND IMPLEMENTATION OF INTELLIGENT TRAFFIC MANAGEMENT SYSTEM (ITMS) TO MINIMIZE TRAFFIC CONGESTION**" in partial fulfillment of the requirements for the award of the degree of Doctor of Philosophy in "**Department of Computer Science & Engineering"** in **Desh Bhagat University**, Mandi Gobindgarh is an authentic record of my own work carried under the supervision of **Prof. (Dr.) Harsh Sadawarti** (Supervisor) and **Dr. Binod Kumar Mishra** (Co-Supervisor).

The matter embodied in this thesis has not been submitted by me for the award of any other degree of this or any other University/Institute.

HASHMAT FIDA
Regd. No. 21RCPHDCS001

This is to certify that the above statement made by the candidate is correct to the best of our knowledge.

Prof. (Dr.) Harsh Sadawarti
(Supervisor)

Dr. Binod Kumar Mishra
(Co-Supervisor)

The Ph.D. Viva-Voce examination of **Hashmat Fida** Research Scholar, has been held on **7th Feb 2025**

Sign. of Supervisor Sign. of Co-Supervisor Sign. of External Examiner

Certificate Regarding Originality of Research Work

Name of Research Scholar: **HASHMAT FIDA**

Name of Department: **Computer Science & Engineering**

Title of the Thesis: **A Framework for Designing and Implementation of Intelligent Traffic Management System (ITMS) To Minimize Traffic Congestion**

Name of Research Supervisor/co-supervisor: **Prof. (Dr.) Harsh Sadawarti**, **Dr. Binod Kumar Mishra**

I / We hereby declare that, the above-mentioned research work is original & it doesn't contain any plagiarized contents. The similarity index of this research work is **2%**.

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DEDICATION

This thesis is dedicated to all those who have inspired, encouraged, and supported me throughout this transformative journey.

To my parents, whose unwavering love, patience, and sacrifices have been the foundation of my perseverance. Your belief in my potential has been my constant source of strength.

To my family and friends, who have been my pillars of support, providing comfort during challenging times and celebrating every milestone along the way.

To my mentors and advisors, whose wisdom, guidance, and encouragement have been instrumental in shaping my academic and professional path. Your dedication to excellence and tireless support have left an indelible mark on me.

To the countless road users whose daily struggles navigating complex traffic systems have fuelled my passion for this work. It is my hope that this research will contribute to safer roads, smarter infrastructure, and a better quality of life for all.

And finally, to the spirit of inquiry and innovation, which reminds us that with persistence and collaboration, we can overcome even the most daunting challenges.

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Above all, I thank **Allah, The Almighty** for the strength and perseverance to complete this journey.

LIST OF ABBREVIATIONS AND SYMBOLS

Abbreviations

- **AI**: Artificial Intelligence
- **CAV**: Connected and Autonomous Vehicles
- **CSI**: Channel State Information
- **DEM**: Digital Elevation Model
- **DL**: Deep Learning
- **DSRC**: Dedicated Short-Range Communications
- **EMV**: Emergency Vehicle
- **GPS**: Global Positioning System
- **ICTWS**: Intersection Cross Traffic Warning System
- **IoT**: Internet of Things
- **ITS**: Intelligent Transportation Systems
- **LiDAR**: Light Detection and Ranging
- **MEMS**: Micro-Electro-Mechanical Systems
- **NLOS**: Non-Line-of-Sight
- **RF**: Radio Frequency
- **SNR**: Signal-to-Noise Ratio
- **STMS**: Smart Traffic Management System
- **TD3**: Twin Delayed Deep Deterministic Policy Gradient
- **V2I**: Vehicle-to-Infrastructure
- **V2X**: Vehicle-to-Everything

Symbols

- d: Distance (meters)
- v: Velocity (meters per second)
- a: Acceleration (meters per second squared)
- t: Time (seconds)
- θ: Angle (degrees or radians)
- λ: Wavelength (micrometers)
- τ: Time delay (seconds)

- σ: Standard deviation
- f: Frequency (Hertz)
- P: Power (Watts)

Unit Abbreviations

- **m**: Meter
- **s**: Second
- **Hz**: Hertz
- **km/h**: Kilometers per hour
- **dB**: Decibels

GLOSSARY

1. **Artificial Intelligence (AI)**: The simulation of human intelligence by machines, enabling them to perform tasks such as learning, reasoning, and decision-making.
2. **Blind Curve**: A road segment with obstructed visibility due to natural or artificial barriers, making it challenging for drivers to see oncoming traffic or hazards.
3. **Connected and Autonomous Vehicles (CAV)**: Vehicles equipped with systems to communicate with infrastructure, other vehicles, or devices, and to operate with minimal or no human intervention.
4. **Digital Elevation Model (DEM)**: A 3D representation of a terrain's surface created from elevation data.
5. **Edge Computing**: A distributed computing paradigm that processes data closer to its source, reducing latency and improving efficiency.
6. **Global Positioning System (GPS)**: A satellite-based navigation system that provides location and time information globally.
7. **Intelligent Transportation Systems (ITS)**: Advanced systems integrating technology and communication to enhance traffic management, safety, and mobility.
8. **Internet of Things (IoT)**: A network of interconnected devices that collect, share, and analyse data for improved functionality and automation.
9. **LiDAR (Light Detection and Ranging)**: A remote sensing technology that uses laser pulses to create detailed 3D maps of an environment.
10. **Micro-Electro-Mechanical Systems (MEMS)**: Miniature mechanical and electrical devices used in sensors and actuators for various applications, including automotive and robotics.
11. **Non-Line-of-Sight (NLOS)**: A scenario in which a direct path between a transmitter and receiver is obstructed, often affecting communication performance.
12. **Radio Frequency (RF)**: The electromagnetic frequency range used for wireless communication, including traffic and vehicle monitoring systems.

13. **Signal-to-Noise Ratio (SNR)**: A measure of signal strength relative to background noise, indicating the quality of a communication link.
14. **Vehicle-to-Everything (V2X)**: Communication technology enabling vehicles to interact with infrastructure, other vehicles, and road users to enhance safety and efficiency.
15. **Vehicle-to-Infrastructure (V2I)**: A subset of V2X communication that allows vehicles to exchange data with road infrastructure, such as traffic lights and signs.
16. **Ultra-Wideband (UWB)**: A wireless communication technology characterized by high bandwidth, used for precise positioning and tracking in traffic systems.
17. **Traffic Congestion**: A condition of slowed or stopped traffic due to high vehicle density, often resulting in delays and inefficiency.
18. **Dynamic Green Corridor**: A system that prioritizes and clears traffic routes for emergency vehicles in real time to minimize response delays.
19. **Point Cloud**: A collection of data points in 3D space generated by sensors like LiDAR, used to map environments and detect obstacles.